Time limit: 50 minutes.
Instructions: This test contains 10 short answer questions. All answers must be expressed in simplest form unless specified otherwise. Only answers written inside the boxes on the answer sheet will be considered for grading.
No calculators.

1. A class of six students has to split into two indistinguishable teams of three people. Compute the number of distinct team arrangements that can result.

2. What is the probability that a randomly chosen factor of 2016 is a perfect square?

3. Compute the remainder when:
$$5\underbrace{6666\ldots6666}_{\text{2016 sixes}}5$$
is divided by 17.

4. At an M&M factory, two types of M&Ms are produced, red and blue. The M&Ms are transported individually on a conveyor belt. Anna is watching the conveyor belt, and has determined that four out of every five red M&Ms are followed by a blue one, while one out of every six blue M&Ms is followed by a red one. What proportion of the M&Ms are red?

5. Three cards are chosen from a standard deck of 52 without replacing them. Given that the ace of spades was chosen, what is the expected number of aces chosen?

6. Moor decides that he needs a new email address, and forms the address by taking some permutation of the 12 letters $MMMOOOOOORRR$. How many permutations of the letters will contain $MOOR$ in this exact order at least once?

7. Suppose that the 8 corners of a cube can be colored either red, green, or blue. We call a coloring of the cube rotationally symmetric if the cube can be rotated along a single axis parallel to an edge of a cube either 90°, 180°, or 270°, and reach the original coloring. How many rotationally symmetric colorings exist using the 3 colors? Assume that any colorings which are identical after rotation are equivalent.

8. Let $x = \frac{1}{9} + \frac{1}{99} + \frac{1}{999} + ... + \frac{1}{999999999}$. Compute the number of digits in the first 3000 decimal places of the base 10 representation of x which are greater than or equal to 8.

9. Two 20-sided dice are rolled. Their outcomes are independent and take uniformly distributed integer values from 1 to 20, inclusive. For each roll, let x be (the sum of the dice) $\times$ (the positive difference of the dice). What is the expected value of x?

10. Compute
$$\sum_{a=1}^{1000}\sum_{b=1}^{1000}\sum_{c=1}^{1000}\left\lfloor\frac{1000}{\operatorname{lcm}(a,b,c)}\right\rfloor\varphi(a)\,\varphi(b)\,\varphi(c)$$
where $\varphi(n) = |\{k : 1 \leq k \leq n, \gcd(k,n) = 1\}|$ counts the integers coprime to n that are less than or equal to n.